PROGRAMMING PROBLEM

ATM Banking System Using Switch-Case in C

Problem Statement

A bank wants to develop a simple ATM Banking System using the C programming language. The ATM should allow customers to check their account balance, deposit money, and withdraw money. Write a C program using a **switch-case** statement to implement the system. Initially, the customer's account balance is **Rs. 5,000**.

ATM Menu

```
========== ATM MENU ==========
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
==============================
Enter your choice:
```

Functional Requirements

1. Check Balance (Option 1)
Display the current account balance.

2. Deposit Money (Option 2)
Ask the user to enter a deposit amount. If the amount is greater than zero, add it to the existing balance and display a success message and updated balance. Otherwise, display **Invalid Amount**.

3. Withdraw Money (Option 3)
Ask the user to enter a withdrawal amount. If the amount is greater than zero and does not exceed the available balance, deduct it and display a success message and remaining balance. If the amount exceeds the available balance, display **Insufficient Balance!**. If the amount is zero or negative, display **Invalid Amount**.

4. Exit (Option 4)
Display **Thank you for using ATM!**.

5. Invalid Choice
For any menu choice other than 1, 2, 3, or 4, display **Invalid Choice!**.

Input Format

• The first input is an integer representing the menu choice.
• For option 2, the next input is the deposit amount (a numeric value).
• For option 3, the next input is the withdrawal amount (a numeric value).
• No additional input is required for options 1 and 4.

Output Format

Display the appropriate result for the selected operation: current balance, deposit confirmation and updated balance, withdrawal confirmation and remaining balance, exit confirmation, or an error message for invalid amounts or choices.

Sample Test Cases

Test 1 – Check Balance

```
Input: 1
Output: Your Balance is: Rs. 5000.00
```

Test 2 – Successful Deposit

```
Input: 2
2000

Output: Deposit Successful!
Updated Balance: Rs. 7000.00
```

Test 3 – Successful Withdrawal

```
Input: 3
1500

Output: Withdrawal Successful!
Remaining Balance: Rs. 3500.00
```

Test 4 – Insufficient Balance

```
Input: 3
6000

Output: Insufficient Balance!
```

Test 5 – Invalid Deposit Amount

```
Input: 2
-500

Output: Invalid Amount
```

Test 6 – Invalid Withdrawal Amount

```
Input: 3
0

Output: Invalid Amount
```

Test 7 – Exit

```
Input: 4

Output: Thank you for using ATM!
```

Test 8 – Invalid Menu Choice

```
Input: 7

Output: Invalid Choice!
```

Constraints

• Initial account balance is Rs. 5,000.
• Menu choice must be an integer.
• Deposit and withdrawal amounts must be numeric and greater than zero.
• A withdrawal cannot exceed the available balance.
• The program performs only one operation per execution; the balance resets to Rs. 5,000 when the program is run again.

Concepts to Be Used

1. Variables and data types
2. printf() and scanf()
3. switch-case statements
4. if-else conditional statements
5. Arithmetic operators
6. Comparison operators

Learning Objective: Students should be able to implement a real-world banking scenario using switch-case, conditional statements, and arithmetic operations in C.